

How to Prove It - Example 2.3.3

Solutions in brief

Part 1

Investigate the logical form of $x \in \mathcal{P} A$.

Solution

$$x \in \mathcal{P} A \equiv (\forall z \mid z \in x : z \in A)$$



Part 2

Investigate the logical form of $\mathcal{P} A \subseteq \mathcal{P} B$.

Solution

$$\mathcal{P} A \subseteq \mathcal{P} B \equiv (\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B))$$



Part 3

Investigate the logical form of $B \in \{\mathcal{P} A \mid A \in \mathcal{F}\}$

Solution

$$B \in \{\mathcal{P} A \mid A \in \mathcal{F}\} \equiv (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid z \in B \equiv (\forall y \mid y \in z : y \in A)))$$



Detailed answers

Each solution was discovered by direct calculation. I used the laws from LADM and organised the logical transformations using the linear proof layout of Back's refinement calculus.

Parts 1 and 2 were straightforward requiring only a couple of definitions to transform the given statement into a quantified term.

Calculation for Part 1.

◆ $x \in \mathcal{P} A$
≡ HTPI (2.3.2) Powerset
 $x \in \{x \mid x \subseteq A\}$
≡ LADM (11.6) Characteristic predicate
 $x \subseteq A$
≡ LADM (11.13) Axiom, Subset
 $(\forall z \mid z \in x : z \in A)$

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Calculation for Part 2.

◆ $\mathcal{P} A \subseteq \mathcal{P} B$
≡ LADM (11.13) Axiom, Subset
 $(\forall X \mid X \subseteq A : X \subseteq B)$
≡ LADM (11.13) Axiom, Subset, twice
 $(\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B))$

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Part 3 was interesting for two reasons. Firstly, I needed to reinterpret traditional notation for set-comprehension in the uniform syntax of LADM. Secondly, I used a sub-calculation to focus on manipulating a subterm, thus demonstrating a streamlined calculation.

Calculation for Part 3.

◆ $B \in \{\mathcal{P} A \mid A \in \mathcal{F}\}$
 ≡ Interpret using LADM notation
 $B \in \{A \mid A \in \mathcal{F} : \mathcal{P} A\}$
 ≡ LADM(11.3) Axiom, Set membership
 $(\exists A \mid A \in \mathcal{F} : B = \mathcal{P} A)$
 ≡ Part 1.
 $(\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : B = \mathcal{P} A)$
 ≡ Focus on term of quantification
 ◆ $B = \mathcal{P} A$
 ≡ LADM (11.4) Axiom, Extensionality
 $(\forall z \mid : z \in B \equiv z \in \mathcal{P} A)$
 ≡ Part 1.
 $(\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A))$
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 ... $(\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A)))$
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I delight in turning quantifier-heavy formulae into pointfree expressions of relation algebra. This exercise is asking us to do the opposite of that. Yuck.